COURSE 2 · SECTION 13 · WORKED SOLUTION

Test types, ppm by product and the cost of a claim

Section 13 · Product quality & reliability dashboard (portfolio project 2)

DESIGN QUESTION

Which validation test type fails most often, is the seat-ring-wear rate the same in every product, and what does a typical claim cost?

Steps

1. Open Engxtech_S13.pbix > File > Save as > Engxtech_S13_Exercise.
2. Table: TestResults[TestType], Lots Tested, Validation Pass Rate; sort by the rate ascending.
3. Table: Products[ProductID], Wear Failures 90d, Matured Units, Field ppm, CI Low ppm, CI High ppm.
4. Load BOM (sheet) if you want to read QtyPer from the data, or type the three values; write Seat Ring ppm (below) and add it to the product table.
5. Measures Avg Seat Claim Cost and Other Warranty Cost (below); cards with Display units None.
6. Measure Lines RPN After 100+ (below); filter the DFMEA table to RPN After >= 100 to name the lines.

DAX new in this exercise

```
Seat Rings Fitted =
SUMX ( FILTER ( Builds, Builds[Matured] ),
    Builds[UnitsBuilt] * LOOKUPVALUE ( BOM[QtyPer], BOM[ProductID], Builds[ProductID], BOM[PartID], "P300" ) )
Seat Ring ppm = DIVIDE ( [Wear Failures 90d], [Seat Rings Fitted] ) * 1000000
Avg Seat Claim Cost = DIVIDE ( [Seat Ring Warranty Cost], CALCULATE ( [Claims], WarrantyClaims[IsSeatRingWear] =
TRUE () ) )
Other Warranty Cost = CALCULATE ( [Warranty Cost], WarrantyClaims[IsSeatRingWear] = FALSE () )
Lines RPN After 100+ = COUNTROWS ( FILTER ( DFMEA, DFMEA[RPN After] >= 100 ) )
```

Endurance is lowest at 89.8% (44 of 49), but the three test types are within a few points of each other on 49 to 66 lots, so this is no clear difference. Per product unit, the two pumps are near 3,200 ppm and the valve near 1,000. Part of the gap is the BOM: a pump has two seat rings and the valve one. Per seat ring, the pumps drop to about 1,600 and the valve stays at 1,018, with only 7 failures and an interval of 409 to 2,097, so treat the valve's lower rate with care.

The ppm counts only seat-ring-wear claims that failed within 90 days of a matured build month: 93. Of the 207 claims, 110 failed after 90 days and 4 come from build months that are not yet matured. Warranty cost counts all of them, which is why cost and ppm answer different questions.

The two DFMEA lines still at RPN 100 or more after actions are the two high-severity lines with no action, DF-04 and DF-07.

Expected result

Check	Result
Pass rate, Dimensional	**91.8% (45 of 49)**
Pass rate, Endurance	**89.8% (44 of 49)**
Pass rate, Environmental	**90.9% (60 of 66)**
CP-100: failures / units / ppm (95 % CI)	**50 / 15,726 / 3,179 (2,360 to 4,192)**
BP-200: failures / units / ppm (95 % CI)	**36 / 11,291 / 3,188 (2,233 to 4,414)**
MV-300: failures / units / ppm (95 % CI)	**7 / 6,877 / 1,018 (409 to 2,097)**
ppm per seat ring: CP-100 / BP-200 / MV-300	**1,590 / 1,594 / 1,018**
Average seat-ring-wear claim cost (USD)	**426.31**
Warranty cost, other failure modes (USD)	**6,530**
DFMEA lines with RPN After >= 100	**2 (DF-04, DF-07)**
Seat-ring-wear claims: in ppm / after 90 d / unmatured month	**93 / 110 / 4**